

Statistics II — Formula Sheet

Bachelor's Degree in Management · Academic Year 2025/2026

Probability Models

Discrete Probability Models

Binomial Distribution

$$X \sim \text{Binomial}(n, p)$$

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}, \quad x = 0, 1, \dots, n$$

$$E(X) = np \quad V(X) = npq \quad q = 1 - p$$

Hypergeometric Distribution

$$X \sim \text{Hypergeometric}(N, M, n) \quad p = \frac{M}{N}$$

$$P(X = x) = \frac{\binom{M}{x} \binom{N-M}{n-x}}{\binom{N}{n}}, \quad \max(0, M + n - N) \leq x \leq \min(M, n)$$

$$E(X) = np \quad V(X) = npq \frac{N-n}{N-1}$$

Poisson Distribution

$$X \sim \text{Poisson}(\lambda), \quad \lambda > 0$$

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0, 1, 2, \dots$$

$$E(X) = V(X) = \lambda$$

Geometric Distribution

$$X \sim \text{Geometric}(p), \quad 0 \leq p \leq 1$$

$$f(x) = p(1-p)^{x-1}, \quad x = 1, 2, \dots \quad F(x) = 1 - (1-p)^k, \quad k \leq x < k+1$$

$$E(X) = \frac{1}{p} \quad V(X) = \frac{1-p}{p^2}$$

Continuous Probability Models

Uniform Distribution

$$X \sim \text{Uniform}(a, b), \quad a, b \in \mathbb{R}, \quad a < b$$

$$F(x) = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & a \leq x < b \\ 1 & x \geq b \end{cases}$$

$$E(X) = \frac{a+b}{2} \quad V(X) = \frac{(b-a)^2}{12}$$

Exponential Distribution

$$X \sim \text{Exponential}(\lambda), \quad \lambda > 0$$

$$F(x) = \begin{cases} 0 & x \leq 0 \\ 1 - e^{-\lambda x} & x > 0 \end{cases}$$

$$E(X) = \frac{1}{\lambda} \quad V(X) = \frac{1}{\lambda^2}$$

Normal Distribution

$$X \sim \text{Normal}(\mu, \sigma), \quad \mu \in \mathbb{R}, \sigma > 0$$

$$E(X) = \mu \quad V(X) = \sigma^2$$

$$Z = \frac{X - \mu}{\sigma} \sim \text{Normal}(0, 1) \quad E(Z) = 0 \quad V(Z) = 1$$

$$\Phi(z) = P(Z \leq z) \quad \Phi(-z) = 1 - \Phi(z)$$

t-Student Distribution

$$T \sim t_{(n)}, \quad n \text{ degrees of freedom}$$

$$E(T) = 0 \quad V(T) = \frac{n}{n-2}, \quad n > 2$$

Chi-Square Distribution

$$Q \sim \chi_{(n)}^2, \quad n \text{ degrees of freedom}$$

$$E(Q) = n \quad V(Q) = 2n$$

$$Y = \sum_{i=1}^n Z_i^2 \sim \chi_{(n)}^2, \quad Z_i \sim \text{Normal}(0, 1) \text{ i.i.d.}$$

Sampling

Sample Statistics

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

$$S^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{1}{n} \sum X_i^2 - \bar{X}^2$$

$$S'^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{1}{n-1} \sum X_i^2 - \frac{n}{n-1} \bar{X}^2$$

$$S^2 = \frac{n-1}{n} S'^2 \quad S'^2 = \frac{n}{n-1} S^2$$

Sampling Distributions and Confidence Intervals for μ

#	Population	Distribution of $\bar{X}$	$(1-\alpha) \times 100\%$ Confidence Interval
1	$X \sim N(\mu, \sigma); \sigma^2$ known	$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$	$\left(\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$
2	$X \sim N(\mu, \sigma); \sigma^2$ unknown	$T = \frac{\bar{X} - \mu}{S'/\sqrt{n}} \sim t_{(n-1)}$	$\left(\bar{x} - t_{\alpha/2} \frac{s'}{\sqrt{n}}, \bar{x} + t_{\alpha/2} \frac{s'}{\sqrt{n}} \right)$
2b	$X \sim N(\mu, \sigma); \sigma^2$ unknown, $n \geq 30$	$Z = \frac{\bar{X} - \mu}{S'/\sqrt{n}} \sim N(0, 1)$	$\approx \left(\bar{x} - z_{\alpha/2} \frac{s'}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{s'}{\sqrt{n}} \right)$
3a	$X \sim ?; \sigma^2$ known, $n \geq 30$	$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$	$\approx \left(\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$
3b	$X \sim ?; \sigma^2$ unknown, $n \geq 30$	$Z = \frac{\bar{X} - \mu}{S'/\sqrt{n}} \sim N(0, 1)$	$\approx \left(\bar{x} - z_{\alpha/2} \frac{s'}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{s'}{\sqrt{n}} \right)$

Sampling Distribution and Confidence Interval for p

Population	Distribution of $\hat{p} = X/n$	$(1-\alpha) \times 100\%$ Confidence Interval
$X \sim \text{Binomial}(n, p), n \geq 30$	$\frac{\hat{p} - p}{\sqrt{\hat{p}(1-\hat{p})/n}} \sim N(0, 1)$	$\left(\hat{p} - z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}, \hat{p} + z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \right)$ (approx.)

Sampling Distribution and Confidence Interval for σ^2

Population	Distribution of S'^2	$(1-\alpha) \times 100\%$ Confidence Interval
$X \sim N(\mu, \sigma)$	$Q = \frac{(n-1)S'^2}{\sigma^2} \sim \chi^2_{(n-1)}$	$\left(\frac{(n-1)s'^2}{q_{\sup}}, \frac{(n-1)s'^2}{q_{\inf}} \right)$

$$P(Q > q_{\inf}) = 1 - \frac{\alpha}{2} \quad P(Q > q_{\sup}) = \frac{\alpha}{2}$$

Parametric Hypothesis Tests (at level α)

Hypotheses	Test Statistic (TS)	Critical Region
$H_0 : \theta = \theta_0$ vs $H_1 : \theta \neq \theta_0$	TS_0	$(-\infty, q_{\alpha/2}) \cup (q_{1-\alpha/2}, +\infty)$
$H_0 : \theta \leq \theta_0$ vs $H_1 : \theta > \theta_0$	TS_0	$(q_{1-\alpha}, +\infty)$
$H_0 : \theta \geq \theta_0$ vs $H_1 : \theta < \theta_0$	TS_0	$(-\infty, q_{\alpha})$

$$P(TS > q_{\alpha}) = \alpha \quad \text{Reject } H_0 \text{ if } \alpha > p\text{-value}$$

$$\alpha = P(\text{Reject } H_0 \mid H_0 \text{ true}) \quad (\text{Type I error probability})$$